

The Program in Action

When you run *fastdraw.py*, the output will look like this:

```
Fast Draw, by Al Sweigart al@inventwithpython.com
```

```
Time to test your reflexes and see if you are the fastest
draw in the west!
When you see "DRAW", you have 0.3 seconds to press Enter.
But you lose if you press Enter before "DRAW" appears.
```

```
Press Enter to begin...
```

```
It is high noon...
DRAW!
```

```
You took 0.3485 seconds to draw. Too slow!
Enter QUIT to stop, or press Enter to play again.
> quit
Thanks for playing!
```

How It Works

The `input()` function pauses the program while waiting for the user to enter a string. This simple behavior means that we can't create real-time games with just `input()`. However, your programs will *buffer* keyboard input, meaning that if you pressed the C, A, and T keys before `input()` is called, those characters will be saved, and they'll appear immediately once `input()` executes.

By recording the time just before the `input()` call on line 22 and the time just after the `input()` call on line 24, we can determine how long the player took to press ENTER. However, if they pressed ENTER before `input()` was called, the buffered ENTER press causes `input()` to return immediately (usually in about 3 milliseconds). This is why line 26 checks if the time was less than 0.01 seconds, or 10 milliseconds, to determine that the player pressed ENTER too soon.

```
1. """Fast Draw, by Al Sweigart al@inventwithpython.com
2. Test your reflexes to see if you're the fastest draw in the west.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, beginner, game"""
5.
6. import random, sys, time
7.
8. print('Fast Draw, by Al Sweigart al@inventwithpython.com')
9. print()
10. print('Time to test your reflexes and see if you are the fastest')
11. print('draw in the west!')
12. print('When you see "DRAW", you have 0.3 seconds to press Enter.')
13. print('But you lose if you press Enter before "DRAW" appears.')
14. print()
15. input('Press Enter to begin...')
```